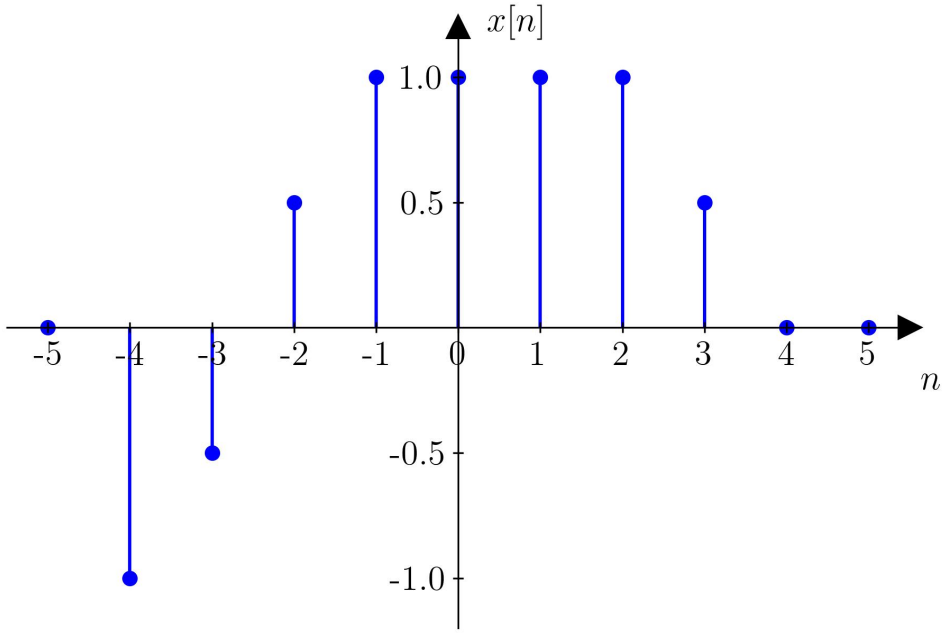
# Exercícios para Praticar

## Processamento de Sinais

**1)** Considere o sinal $x[n]$ mostrado abaixo.




Para este sinal, esboce o gráfico de cada um dos seguintes sinais (lembrete: $u[n]$ e $\delta[n]$ denotam, respectivamente, os sinais degrau unitário e impulso unitário):

a) $x[n-4]$

b) $x[3-n]$

c) $x[3n]$

d) $x[3n+1]$

e) $x[n] \cdot u[3-n]$

f) $x[n-2] \cdot \delta[n-2]$

**2)** Esboce o gráfico de cada um dos sinais abaixo:

a) $x[n] = u[n] - u[n-4]$

b) $x[n] = u[3-n]$

c) $x[n] = -2\delta[n+2] + 2\delta[n-2]$

d) $x[n] = \left(\frac{1}{2}\right)^n \cdot u[n]$

e) $x[n] = \cos(\pi n) \cdot (u[n+2] - u[n-2])$

f) $x[n] = \cos\left(\frac{\pi n}{2}\right) u[n]$

**3)** Para cada um dos sinais de tempo discreto a seguir, determine se eles são periódicos e, em casos afirmativos, encontre o seu período fundamental:

a) $x[n] = \cos\left(\frac{\pi n}{6}\right)$

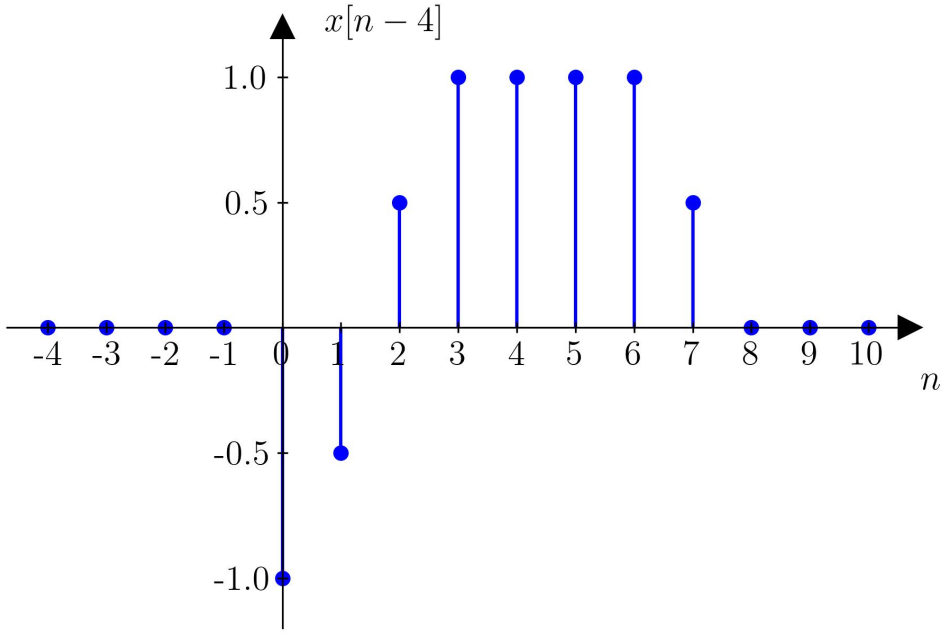
b) $x[n] = \frac{\operatorname{sen}\left(\frac{\pi n}{4}\right)}{\pi n}$

c) $x[n] = \cos\left(\frac{\pi n}{\sqrt{2}}\right)$

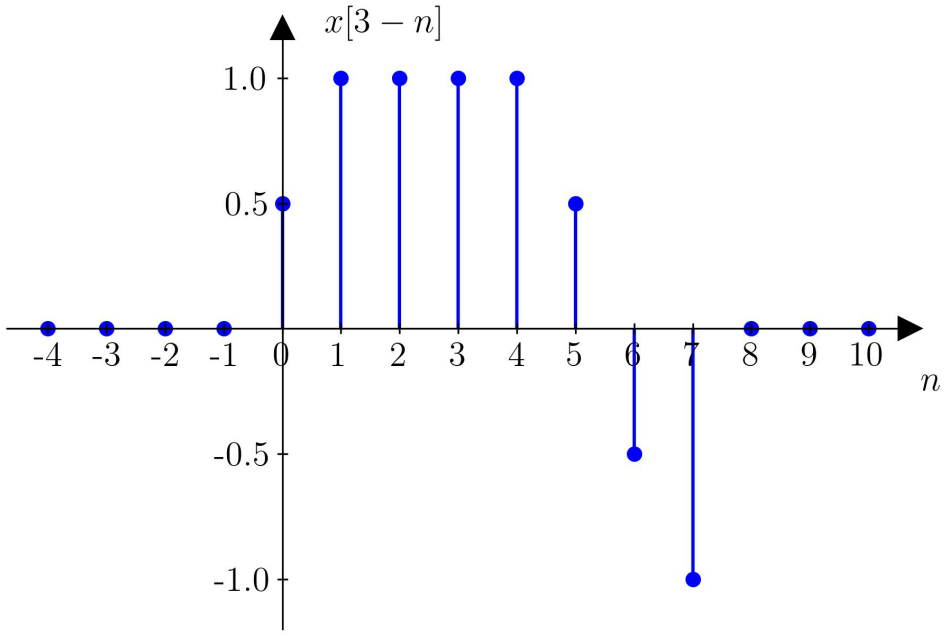
c) $x[n] = \operatorname{sen}\left(\frac{6\pi n}{7} + 1\right)$

**Respostas**
**1)**

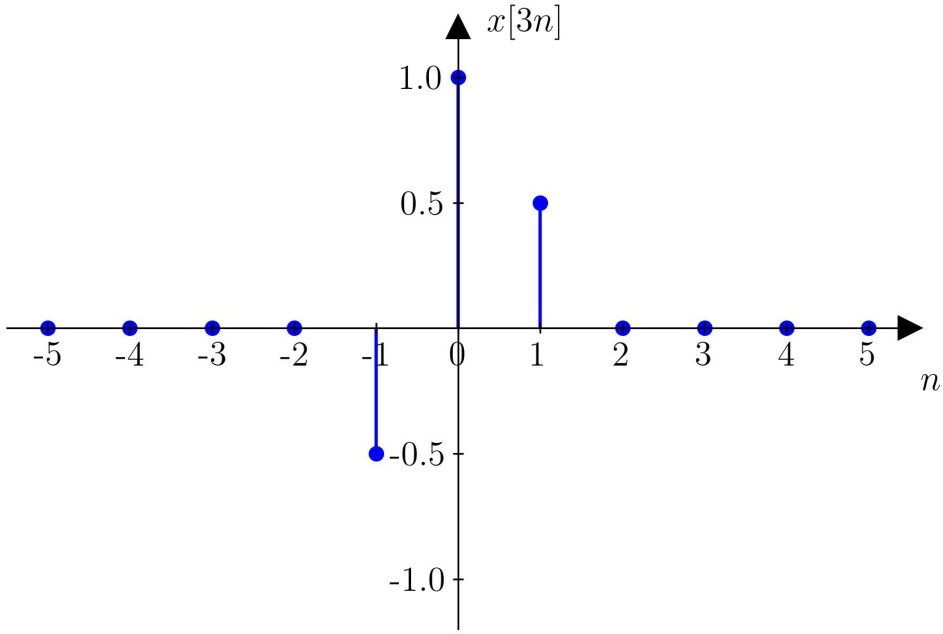
a)



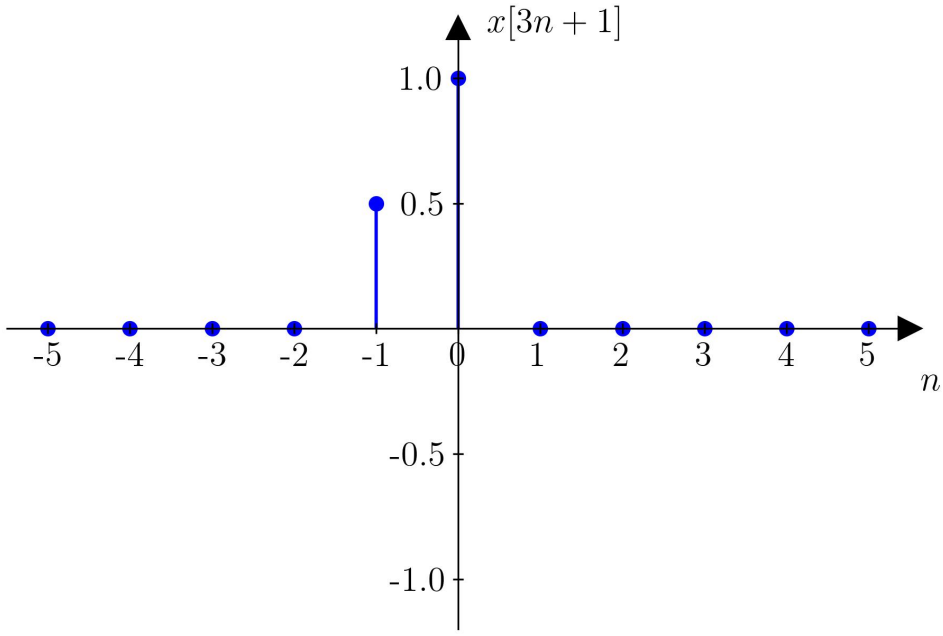

b)



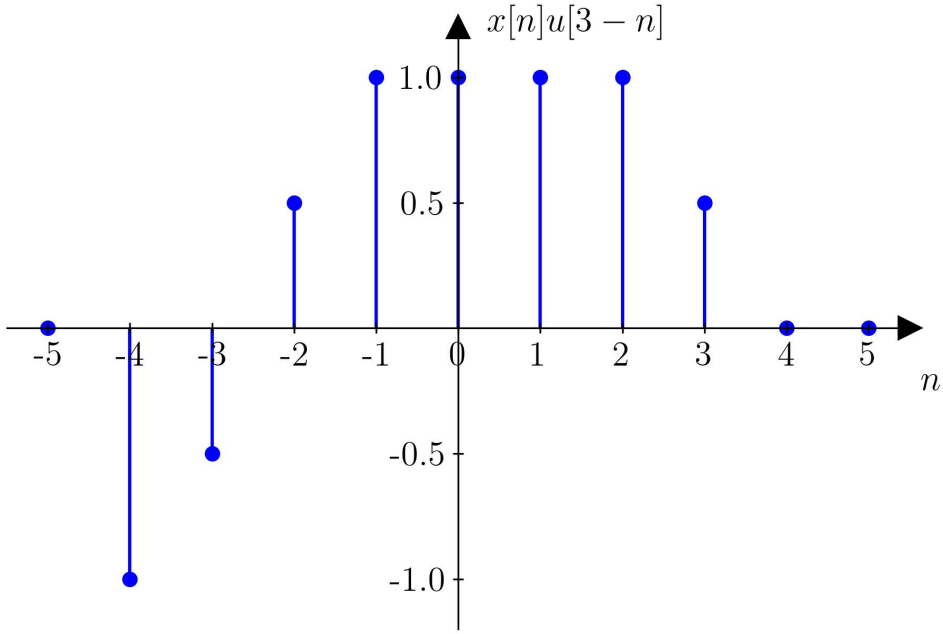
c)



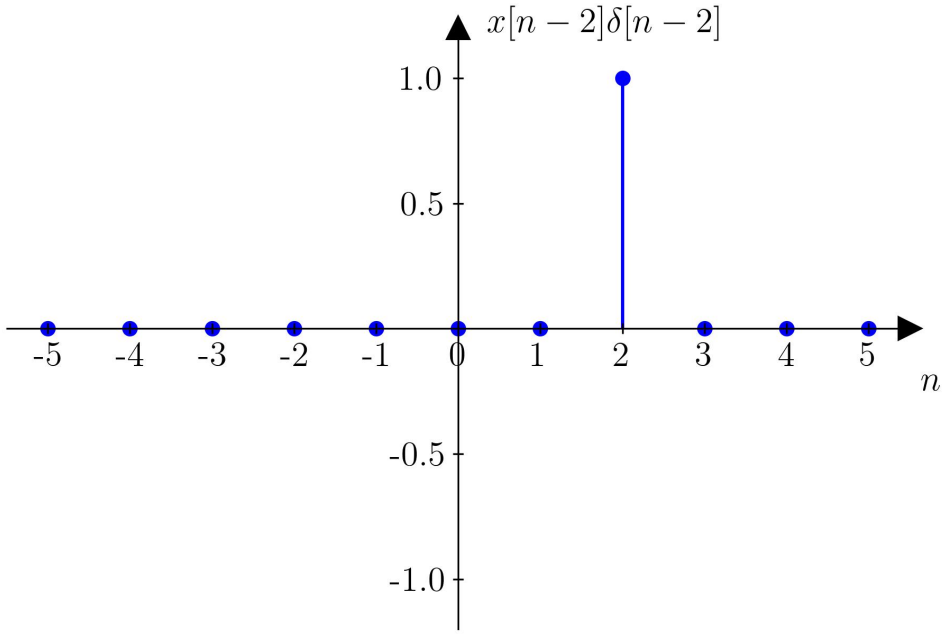

d)



e)

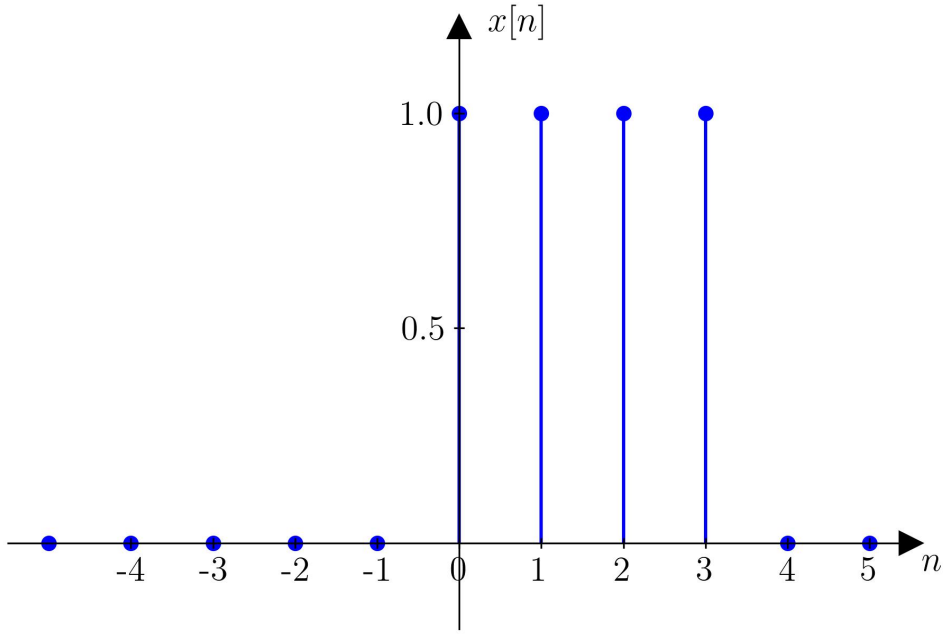

$x[n]u[3-n]$
1.0
0.5
-0.5
-1.0
-5
-4
-3
-2
-1
0
1
2
3
4
5
$n$

f)

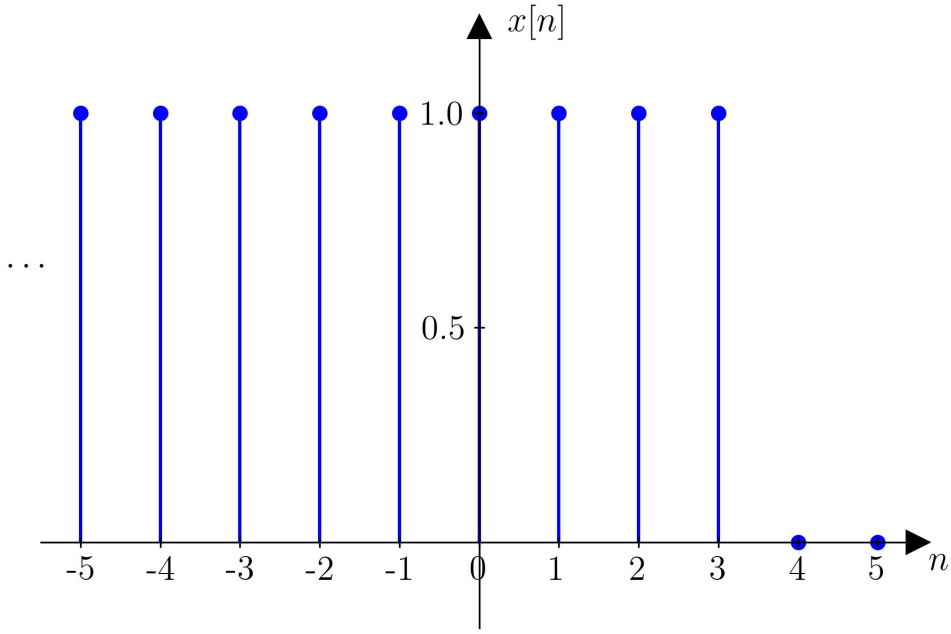

$x[n-2]\delta[n-2]$
1.0
0.5
-0.5
-1.0
-5
-4
-3
-2
-1
0
1
2
3
4
5
$n$

**2)**

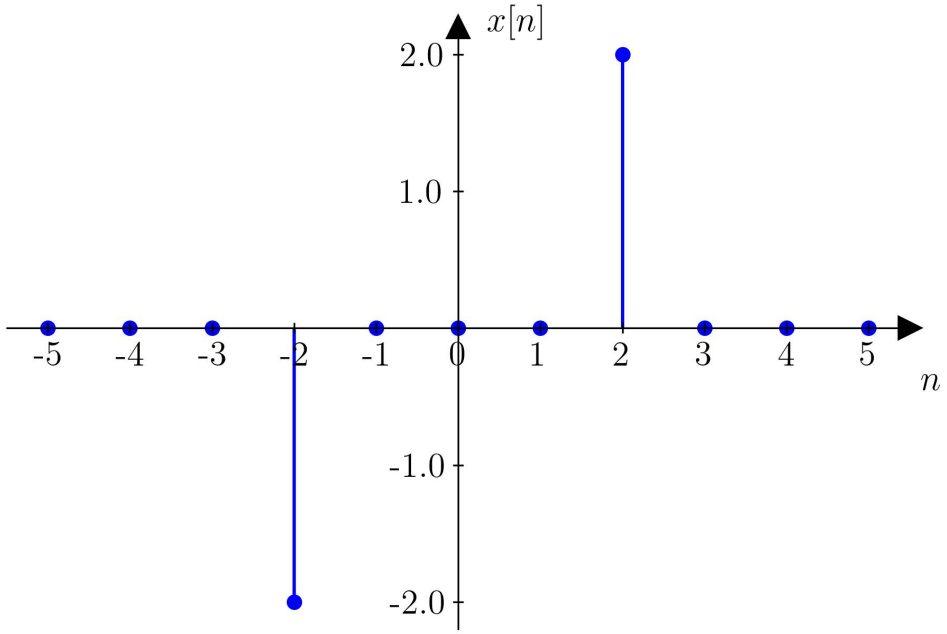
a)



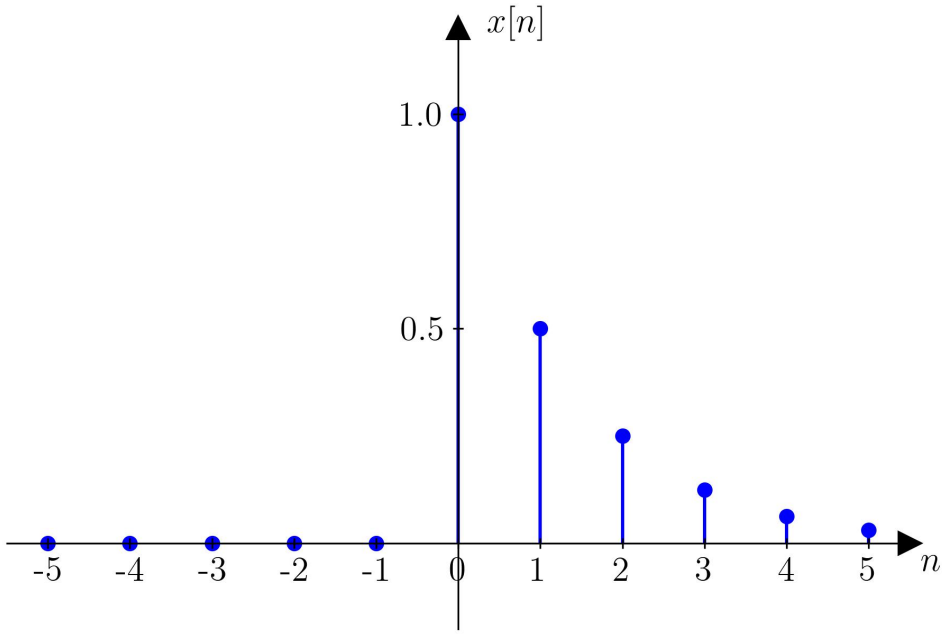

b)



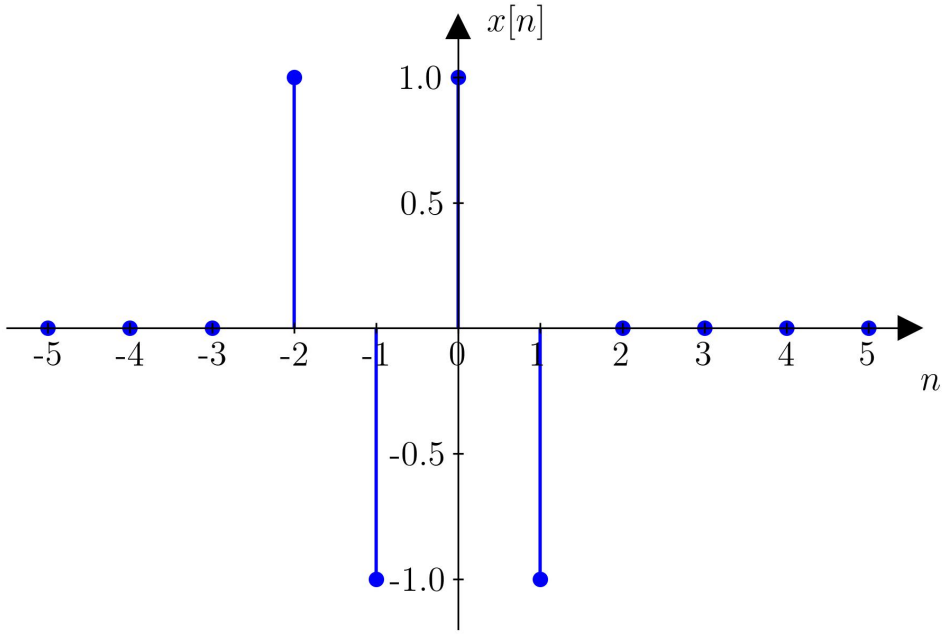
c)



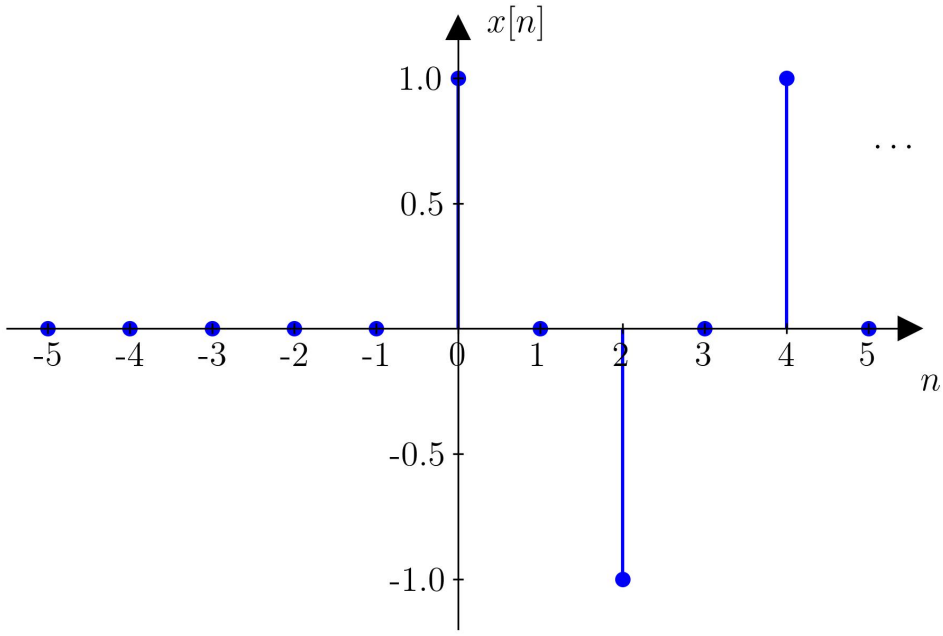

d)



e)




f)



**3)**

a) Sim, com período fundamental $N_0 = 12$

b) Não, o sinal **não** é periódico

c) Não, o sinal **não** é periódico

c) Sim, com período fundamental $N_0 = 7$